PART1: Answer Questions

1. **When ONOS activate "org.onosproject.openflow" , what APPs does it activate?**

```
demo@root > app deactivate org.onosproject.openflow
Deactivated org.onosproject.openflow
demo@root > apps -a -s
*  28 org.onosproject.hostprovider          2.7.0    Host Location Provider
*  61 org.onosproject.drivers               2.7.0    Default Drivers
* 130 org.onosproject.gui2                  2.7.0    ONOS GUI2
demo@root > app activate org.onosproject.openflow
Activated org.onosproject.openflow
demo@root > apps -a -s
*   4 org.onosproject.optical-model         2.7.0    Optical Network Model
*  28 org.onosproject.hostprovider          2.7.0    Host Location Provider
*  30 org.onosproject.openflow-base         2.7.0    OpenFlow Base Provider
*  42 org.onosproject.lldpprovider          2.7.0    LLDP Link Provider
*  43 org.onosproject.openflow              2.7.0    OpenFlow Provider Suite
*  61 org.onosproject.drivers               2.7.0    Default Drivers
* 130 org.onosproject.gui2                  2.7.0    ONOS GUI2
```

如圖所示 "org.onosproject.openflow" 會啟用 4 個 apps:

org.onosproject.optical-model

org.onosproject.openflow-base

org.onosproject.lldpprovider

org.onosproject.openflow

2. **After we activate ONOS and run P.17 Mininet command, will H1 ping H2 successfully? Why or why not?**

```
mininet> h1 ping -c3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
From 10.0.0.1 icmp_seq=1 Destination Host Unreachable
From 10.0.0.1 icmp_seq=2 Destination Host Unreachable
From 10.0.0.1 icmp_seq=3 Destination Host Unreachable

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2084ms
```

如圖所示，嘗試進行 h1 ping h2 後失敗了。原因可以從 Basic ONOS Tutorial 中找到，"There are no flows installed on the data-plane, which forward the traffic appropriately. ONOS comes with a simple Reactive Forwarding app that installs forwarding flows on demand, but this application is not activated by default."

```
mininet> h1 ping -c3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=43.6 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.986 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.078 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2012ms
rtt min/avg/max/mdev = 0.078/14.871/43.551/20.282 ms
```

在啟用 Reactive Forwarding app 之後，就可以順利完成 h1 ping h2 的操作了。

3. **Which TCP port does the controller listen to the OpenFlow connection request from the switch? (Take screenshot and explain your answer.)**

```
demo@root > apps -a -s
*  28 org.onosproject.hostprovider         2.7.0    Host Location Provider
*  61 org.onosproject.drivers              2.7.0    Default Drivers
* 130 org.onosproject.gui2                 2.7.0    ONOS GUI2
demo@root >
demo@SDN-NFV:~/onos$ sudo netstat -nlpt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      774/sshd: /usr/sbin
tcp        0      0 127.0.0.1:5005          0.0.0.0:*               LISTEN      3015/java
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      742/cupsd
tcp        0      0 0.0.0.0:6656            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      478/systemd-resolve
tcp        0      0 0.0.0.0:6655            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 0.0.0.0:6654            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp6       0      0 :::37475                :::*                    LISTEN      3015/java
tcp6       0      0 :::22                   :::*                    LISTEN      774/sshd: /usr/sbin
tcp6       0      0 :::9876                 :::*                    LISTEN      3015/java
tcp6       0      0 :::1099                 :::*                    LISTEN      3015/java
tcp6       0      0 127.0.0.1:45041         :::*                    LISTEN      3015/java
tcp6       0      0 ::1:37885               :::*                    LISTEN      2656/bazel(onos)
tcp6       0      0 ::1:631                 :::*                    LISTEN      742/cupsd
tcp6       0      0 :::8101                 :::*                    LISTEN      3015/java
tcp6       0      0 :::8181                 :::*                    LISTEN      3015/java
```

```
demo@root > apps -a -s
*   4 org.onosproject.optical-model        2.7.0    Optical Network Model
*  28 org.onosproject.hostprovider         2.7.0    Host Location Provider
*  30 org.onosproject.openflow-base        2.7.0    OpenFlow Base Provider
*  34 org.onosproject.fwd                  2.7.0    Reactive Forwarding
*  42 org.onosproject.lldpprovider         2.7.0    LLDP Link Provider
*  43 org.onosproject.openflow             2.7.0    OpenFlow Provider Suite
*  61 org.onosproject.drivers              2.7.0    Default Drivers
* 130 org.onosproject.gui2                 2.7.0    ONOS GUI2
demo@root >
demo@SDN-NFV:~/onos$ sudo netstat -nlpt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      774/sshd: /usr/sbin
tcp        0      0 127.0.0.1:5005          0.0.0.0:*               LISTEN      3015/java
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      742/cupsd
tcp        0      0 0.0.0.0:6656            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      478/systemd-resolve
tcp        0      0 0.0.0.0:6655            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 0.0.0.0:6654            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp6       0      0 :::37475                :::*                    LISTEN      3015/java
tcp6       0      0 :::22                   :::*                    LISTEN      774/sshd: /usr/sbin
tcp6       0      0 :::9876                 :::*                    LISTEN      3015/java
tcp6       0      0 :::1099                 :::*                    LISTEN      3015/java
tcp6       0      0 127.0.0.1:45041         :::*                    LISTEN      3015/java
tcp6       0      0 ::1:37885               :::*                    LISTEN      2656/bazel(onos)
tcp6       0      0 ::1:631                 :::*                    LISTEN      742/cupsd
tcp6       0      0 :::6653                 :::*                    LISTEN      3015/java
tcp6       0      0 :::6633                 :::*                    LISTEN      3015/java
tcp6       0      0 :::8101                 :::*                    LISTEN      3015/java
tcp6       0      0 :::8181                 :::*                    LISTEN      3015/java
```

在啟用 org.onosproject.openflow 後比對差別，可以發現差別在 6653 和

6633(舊版所用)兩個 port，其中 6653 就是 Port which OpenFlow controller listening for switch。

4. **In question 3, which APP enables the controller to listen on the TCP port?**

```
demo@root > app activate org.onosproject.openflow-base
Activated org.onosproject.openflow-base
demo@root >
demo@SDN-NFV:~/onos$ sudo netstat -nlpt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      774/sshd: /usr/sbin
tcp        0      0 127.0.0.1:5005          0.0.0.0:*               LISTEN      3015/java
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      742/cupsd
tcp        0      0 0.0.0.0:6656            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      478/systemd-resolve
tcp        0      0 0.0.0.0:6655            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 0.0.0.0:6654            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp6       0      0 :::37475                :::*                    LISTEN      3015/java
tcp6       0      0 :::22                   :::*                    LISTEN      774/sshd: /usr/sbin
tcp6       0      0 :::9876                 :::*                    LISTEN      3015/java
tcp6       0      0 :::1099                 :::*                    LISTEN      3015/java
tcp6       0      0 127.0.0.1:45041         :::*                    LISTEN      3015/java
tcp6       0      0 ::1:37885               :::*                    LISTEN      2656/bazel(onos)
tcp6       0      0 ::1:631                 :::*                    LISTEN      742/cupsd
tcp6       0      0 :::6653                 :::*                    LISTEN      3015/java
tcp6       0      0 :::6633                 :::*                    LISTEN      3015/java
tcp6       0      0 :::8101                 :::*                    LISTEN      3015/java
tcp6       0      0 :::8181                 :::*                    LISTEN      3015/java
```

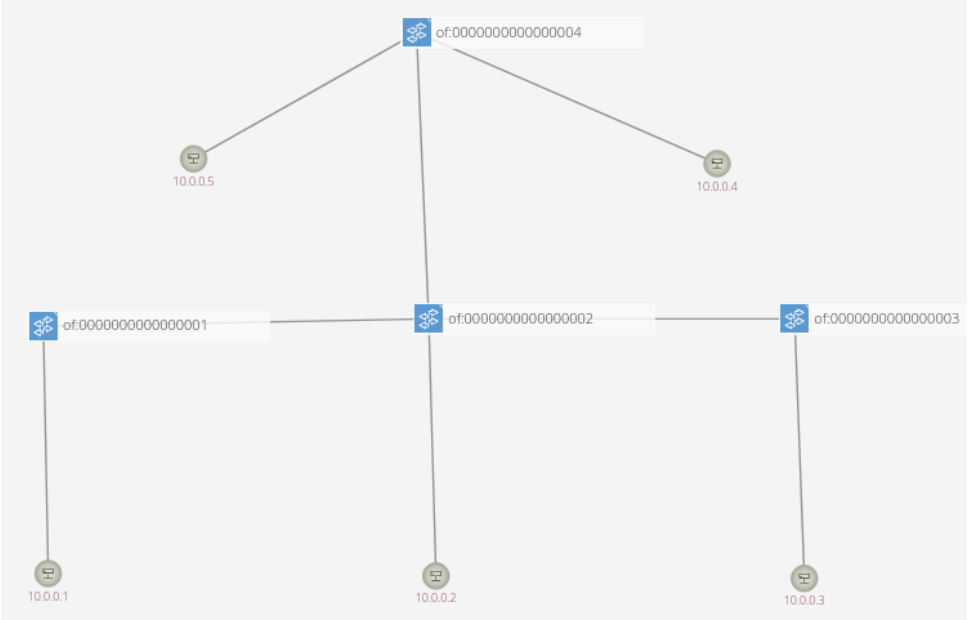
```
demo@root > app deactivate org.onosproject.openflow-base
Deactivated org.onosproject.openflow-base
demo@root >
demo@SDN-NFV:~/onos$ sudo netstat -nlpt
[sudo] password for demo:
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      774/sshd: /usr/sbin
tcp        0      0 127.0.0.1:5005          0.0.0.0:*               LISTEN      3015/java
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      742/cupsd
tcp        0      0 0.0.0.0:6656            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      478/systemd-resolve
tcp        0      0 0.0.0.0:6655            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp        0      0 0.0.0.0:6654            0.0.0.0:*               LISTEN      920/ovs-vswitchd
tcp6       0      0 :::37475                :::*                    LISTEN      3015/java
tcp6       0      0 :::22                   :::*                    LISTEN      774/sshd: /usr/sbin
tcp6       0      0 :::9876                 :::*                    LISTEN      3015/java
tcp6       0      0 :::1099                 :::*                    LISTEN      3015/java
tcp6       0      0 127.0.0.1:45041         :::*                    LISTEN      3015/java
tcp6       0      0 ::1:37885               :::*                    LISTEN      2656/bazel(onos)
tcp6       0      0 ::1:631                 :::*                    LISTEN      742/cupsd
tcp6       0      0 :::8101                 :::*                    LISTEN      3015/java
tcp6       0      0 :::8181                 :::*                    LISTEN      3015/java
```

可以發現是 OpenFlow Base Provider 所造成的影響。

PART2: Create a custom Topology

根據題目需要建立 s1, s2, s3, s4 四個 switch 以及 h1, h2, h3, h4, h5 五個 host 並將他們依據對應關係建立 link

```python
class Lab1_Topo_313551124( Topo ):
    def __init__( self ):
        Topo.__init__( self )

        # Add hosts
        h1 = self.addHost( 'h1' )
        h2 = self.addHost( 'h2' )
        h3 = self.addHost( 'h3' )
        h4 = self.addHost( 'h4' )
        h5 = self.addHost( 'h5' )

        # Add switches
        s1 = self.addSwitch( 's1' )
        s2 = self.addSwitch( 's2' )
        s3 = self.addSwitch( 's3' )
        s4 = self.addSwitch( 's4' )

        # Add links
        self.addLink( h1, s1 )
        self.addLink( h2, s2 )
        self.addLink( h3, s3 )
        self.addLink( h4, s4 )
        self.addLink( h5, s4 )

        self.addLink( s1, s2 )
        self.addLink( s2, s3 )
        self.addLink( s2, s4 )


topos = { 'topo_part2_313551124': Lab1_Topo_313551124 }
```



PART3

根據題目所需，手動設置各個 host 的 ip，並將其設置為 27bit 來符合 subnet mask 255.255.255.224

```python
class Lab1_Topo_313551124( Topo ):
    def __init__( self ):
        Topo.__init__( self )

        # Add hosts
        h1 = self.addHost( 'h1' , ip = '192.168.0.1/27')
        h2 = self.addHost( 'h2' , ip = '192.168.0.2/27')
        h3 = self.addHost( 'h3' , ip = '192.168.0.3/27')
        h4 = self.addHost( 'h4' , ip = '192.168.0.4/27')
        h5 = self.addHost( 'h5' , ip = '192.168.0.5/27')

        # Add switches
        s1 = self.addSwitch( 's1' )
        s2 = self.addSwitch( 's2' )
        s3 = self.addSwitch( 's3' )
        s4 = self.addSwitch( 's4' )

        # Add links
        self.addLink( h1, s1 )
        self.addLink( h2, s2 )
        self.addLink( h3, s3 )
        self.addLink( h4, s4 )
        self.addLink( h5, s4 )

        self.addLink( s1, s2 )
        self.addLink( s2, s3 )
        self.addLink( s2, s4 )
```

```
mininet> dump
<Host h1: h1-eth0:192.168.0.1 pid=6075>
<Host h2: h2-eth0:192.168.0.2 pid=6077>
<Host h3: h3-eth0:192.168.0.3 pid=6079>
<Host h4: h4-eth0:192.168.0.4 pid=6081>
<Host h5: h5-eth0:192.168.0.5 pid=6083>
```

```
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.1  netmask 255.255.255.224  broadcast 192.168.0.31
        inet6 fe80::9cb1:65ff:fefd:4d29  prefixlen 64  scopeid 0x20<link>
        ether 9e:b1:65:fd:4d:29  txqueuelen 1000  (Ethernet)
        RX packets 73  bytes 8074 (8.0 KB)
        RX errors 0  dropped 28  overruns 0  frame 0
        TX packets 25  bytes 1846 (1.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.2  netmask 255.255.255.224  broadcast 192.168.0.31
        inet6 fe80::10db:d7ff:fedb:5f56  prefixlen 64  scopeid 0x20<link>
        ether 12:db:d7:db:5f:56  txqueuelen 1000  (Ethernet)
        RX packets 89  bytes 10298 (10.2 KB)
        RX errors 0  dropped 44  overruns 0  frame 0
        TX packets 25  bytes 1846 (1.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
mininet> h3 ifconfig
h3-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.3  netmask 255.255.255.224  broadcast 192.168.0.31
        inet6 fe80::5cbb:43ff:feb4:91bc  prefixlen 64  scopeid 0x20<link>
        ether 5e:bb:43:b4:91:bc  txqueuelen 1000  (Ethernet)
        RX packets 97  bytes 11309 (11.3 KB)
        RX errors 0  dropped 50  overruns 0  frame 0
        TX packets 26  bytes 1916 (1.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
mininet> h4 ifconfig
h4-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.4  netmask 255.255.255.224  broadcast 192.168.0.31
        inet6 fe80::3c76:3aff:feda:f22e  prefixlen 64  scopeid 0x20<link>
        ether 3e:76:3a:da:f2:2e  txqueuelen 1000  (Ethernet)
        RX packets 97  bytes 11309 (11.3 KB)
        RX errors 0  dropped 50  overruns 0  frame 0
        TX packets 26  bytes 1916 (1.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
mininet> h5 ifconfig
h5-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.5  netmask 255.255.255.224  broadcast 192.168.0.31
        inet6 fe80::81a:9ff:fe7f:fb8b  prefixlen 64  scopeid 0x20<link>
        ether 0a:1a:09:7f:fb:8b  txqueuelen 1000  (Ethernet)
        RX packets 99  bytes 11587 (11.5 KB)
        RX errors 0  dropped 52  overruns 0  frame 0
        TX packets 26  bytes 1916 (1.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

What you’ve learned or solved

了解 onos 基礎環境設置，以及一些簡單的 mininet 的操作。